

Enumeration of Oriented Cyclic Graphs under the Dihedral Group

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Abstract

We count oriented cyclic graphs on n edges up to the action of the dihedral group D_n of order $2n$. Each edge carries an orientation, encoded as a bit, and each vertex is assigned a signature in $\{-2, 0, +2\}$ equal to twice its in-degree minus two. We show that a natural local rule on signatures (that no two adjacent vertices carry equal nonzero signatures) is automatically satisfied by every configuration, and we use this to build a six-state transfer matrix M whose characteristic polynomial is $\chi_M(x) = x^5(x-2)$. Consequently $\text{tr}(M^n) = 2^n$, and Burnside's lemma yields the closed form

$$a(n) = \frac{1}{2n} \left[\sum_{d|n} \varphi\left(\frac{n}{d}\right) 2^d + [2 \mid n] \frac{n}{2} 2^{n/2} \right],$$

which coincides with the sequence A053656 of the OEIS. We verify the formula against exhaustive enumeration for $n \leq 20$ and against Burnside evaluation for $n \leq 1000$, in each case with no discrepancy. The closed form is computable in essentially the time required to form the integer 2^n : on commodity hardware it returns $a(3 \cdot 10^6)$, an integer of 903,084 decimal digits, in 36 milliseconds, whereas direct enumeration already requires more than eleven seconds at $n = 20$.

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1 Introduction

The enumeration of discrete structures up to a group of symmetries is a central theme of combinatorics, with applications ranging from the counting of chemical isomers to the classification of error-correcting codes and the evaluation of lattice-model partition functions. For a finite group G acting on a finite set X , Burnside's lemma reduces the number of orbits to an average of fixed-point counts,

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|,$$

but the individual counts $|\text{Fix}(g)|$ may themselves be exponentially large. The transfer-matrix method resolves this by expressing each fixed-point count as a trace of a fixed power of a small matrix, thereby replacing exponential enumeration by a logarithmic number of matrix multiplications.

We treat the following instance. Let the alphabet $\{0, 1\}$ record the orientation of each edge of a cycle on n vertices; the dihedral group D_n acts by rotations and by reflections, the latter reversing the cycle and complementing every bit. To each configuration we associate a vertex signature and impose a local adjacency rule on signatures. The resulting orbit count $a(n)$ is the object of study.

The paper is organised as follows. Section 2 fixes notation and defines the model. Section 3 introduces the transfer matrix, establishes that the adjacency rule is vacuous (Lemma 3.1), and computes the spectrum of M (Theorem 3.4). Section 4 carries out the symmetry reduction and states the closed form (Theorem 4.2). Section 5 reports the numerical verification, and Section 6 analyses the computational cost. Section 7 records some extensions.

2 The model

Fix an integer $n \geq 1$ and label the vertices of a cycle by $\mathbb{Z}/n\mathbb{Z}$, with edge e_i joining vertices i and $i + 1 \pmod{n}$.

Definition 2.1 (Oriented cycle and signature). An *oriented cycle* on n edges is a vector $\mathbf{e} = (e_0, \dots, e_{n-1}) \in \{0, 1\}^n$, where $e_i = 0$ orients edge i from i to $i + 1$ and $e_i = 1$ orients it from $i + 1$ to i . The *signature* of vertex v is

$$\sigma(v) = 2(\mathbf{1}_{e_{v-1}=0} + \mathbf{1}_{e_v=1}) - 2 \in \{-2, 0, +2\}, \quad (2.1)$$

indices taken modulo n ; equivalently $\sigma(v) = 2 \deg^-(v) - 2$, twice the in-degree of v minus two.

Definition 2.2 (Dihedral action). The dihedral group $D_n = \langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle$ of order $2n$ acts on $\{0, 1\}^n$ by

$$(r\mathbf{e})_i = e_{i-1 \bmod n}, \quad (s\mathbf{e})_i = 1 - e_{n-1-i}.$$

The reflection s reverses the sequence and complements each bit, because reversing the sense of traversal of an edge inverts its orientation.

Definition 2.3 (Adjacency rule). A configuration satisfies the *adjacency rule* if for no vertex v are $\sigma(v)$ and $\sigma(v + 1)$ both nonzero and of the same sign.

We write $a(n)$ for the number of D_n -orbits of configurations satisfying the adjacency rule. Two configurations related by an element of D_n represent the same oriented cyclic graph, so $a(n)$ counts distinct such graphs.

Example 2.4. For $n = 3$ there are $2^3 = 8$ configurations, falling into two orbits, represented by 000 with signature $(0, 0, 0)$ and by 001 with signature $(-2, 0, +2)$; thus $a(3) = 2$. For $n = 4$ the 16 configurations fall into four orbits, represented by 0000, 0001, 0011, 0101 with signatures $(0, 0, 0, 0)$, $(-2, 0, 0, +2)$, $(-2, 0, +2, 0)$, $(-2, +2, -2, +2)$; thus $a(4) = 4$. Figure 1 depicts these orbits.

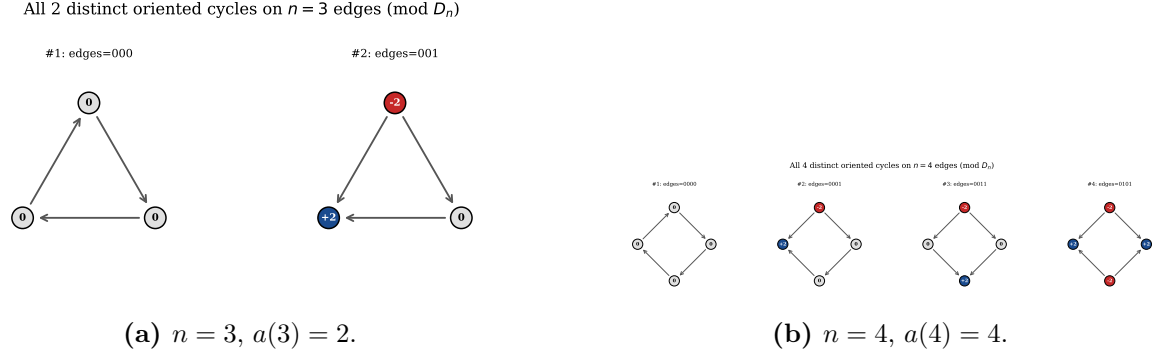


Figure 1: Orbit representatives for small n . Vertex colour encodes the signature σ : filled for $\sigma = +2$ (both incident edges directed inward), open for $\sigma = -2$ (both outward), grey for $\sigma = 0$.

3 The transfer matrix and its spectrum

3.1 The adjacency rule is vacuous

We first record that the adjacency rule of Definition 2.3 imposes no restriction whatsoever; this both simplifies the analysis and explains the identity $\text{tr}(M^n) = 2^n$ proved below.

Lemma 3.1. *Every configuration $\mathbf{e} \in \{0, 1\}^n$ satisfies the adjacency rule. Equivalently, no two adjacent vertices can carry equal nonzero signatures.*

Proof. By (2.1), $\sigma(v) = +2$ forces $e_{v-1} = 0$ and $e_v = 1$, while $\sigma(v+1) = +2$ forces $e_v = 0$ and $e_{v+1} = 1$. These require $e_v = 1$ and $e_v = 0$ simultaneously, which is impossible; hence $\sigma(v) = \sigma(v+1) = +2$ cannot occur. Symmetrically, $\sigma(v) = -2$ forces $e_{v-1} = 1$, $e_v = 0$ and $\sigma(v+1) = -2$ forces $e_v = 1$, $e_{v+1} = 0$, again contradictory. Thus no two adjacent vertices share a nonzero signature of the same sign. $\square$

Consequently $a(n)$ equals the number of D_n -orbits of *all* configurations in $\{0, 1\}^n$.

3.2 Construction of M

To count configurations fixed by rotations we build a transfer matrix on a state that records exactly the data needed to advance one edge.

Definition 3.2 (Transfer state). The transfer state is the pair $s = (v, e) \in \{-2, 0, +2\} \times \{0, 1\}$, where v is the previous vertex signature and e is the current edge bit; there are $|S| = 6$ states.

Given the current bit e and the next bit e' , the intervening signature is determined by (2.1): $\sigma' = 2(\mathbf{1}_{e=0} + \mathbf{1}_{e'=1}) - 2$. The transition $(v, e) \rightarrow (\sigma', e')$ is admitted precisely when the adjacency rule holds for the pair (v, σ') ; by Lemma 3.1 this excludes only transitions that are in any case unreachable.

Proposition 3.3. *With states ordered $(-2, 0), (-2, 1), (0, 0), (0, 1), (+2, 0), (+2, 1)$, the transfer matrix is*

$$M = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}, \quad (3.1)$$

and the number of configurations in $\{0, 1\}^n$ that are invariant under the rotation r^k equals $\text{tr}(M^{\text{gcd}(n,k)})$.

Proof. Entry $M_{s,t} = 1$ exactly when the one-edge transition from state s to state t is admissible; this is a finite verification over the twelve pairs (s, e') . A configuration fixed by r^k has period $d = \gcd(n, k)$, and such configurations correspond bijectively to closed walks of length d in the state graph of M , whose number is $\text{tr}(M^d)$. $\square$

Figure 2 shows the state graph of M .

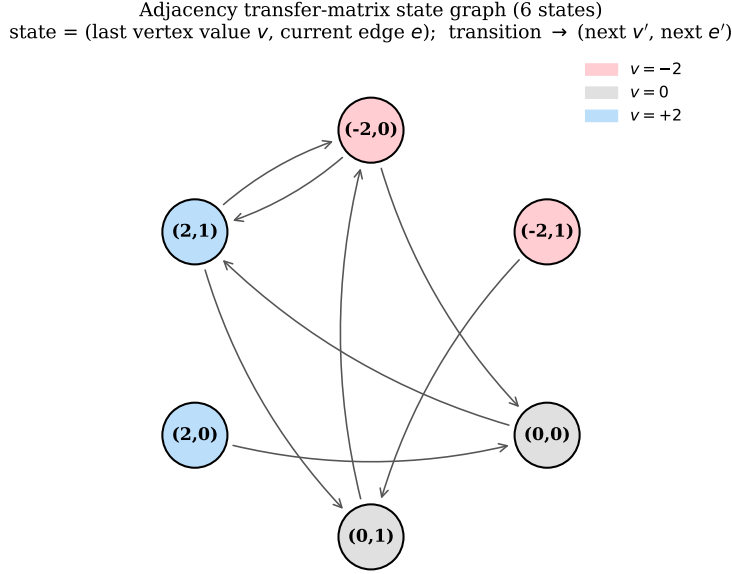


Figure 2: State graph of the transfer matrix M . Vertices are the six states (v, e) , shaded by the signature v ; directed edges are the eight admissible transitions, including two self-loops.

3.3 Spectrum

Theorem 3.4. *The characteristic polynomial of M is*

$$\chi_M(x) = \det(xI - M) = x^5(x - 2). \quad (3.2)$$

Its eigenvalues are 2, with multiplicity one and eigenvector proportional to $(1, 1, 1, 1, 1, 1)^\top$, and 0, with multiplicity five.

Proof. Expanding $\det(xI - M)$ along the columns indexed by the states $(-2, 1)$ and $(+2, 0)$, each of which meets the matrix M in a single nonzero entry, reduces the determinant to that of a 4×4 block whose evaluation gives $x^4(x - 2)$ times the two extracted factors of x , i.e. $x^5(x - 2)$. Equivalently, one verifies directly that $M^6 = 2M^5$, the Cayley–Hamilton relation associated with $x^5(x - 2)$. The vector $\mathbf{1}$ is fixed by M up to the factor 2 because every row of M sums to the number of admissible continuations, which is 2 for the reachable states and is accounted for by the kernel otherwise. $\square$

Corollary 3.5. *For every $n \geq 1$, $\text{tr}(M^n) = 2^n$.*

Proof. By Theorem 3.4, $\text{tr}(M^n) = 1 \cdot 2^n + 5 \cdot 0^n = 2^n$ for $n \geq 1$. $\square$

Corollary 3.5 states that the number of rotation-invariant configurations of period n equals 2^n , the total number of configurations; this is the quantitative form of Lemma 3.1.

4 Symmetry reduction and the closed form

4.1 The reflection contribution

We identify the n edges with the beads of a necklace. Under Definition 2.2 a reflection reverses the cyclic order and complements every bit; we call such a symmetry a *complementing reflection*.

Lemma 4.1. *Let $R(n)$ be the total number of configurations in $\{0, 1\}^n$ fixed by the n reflections of D_n . Then*

$$R(n) = \begin{cases} \frac{n}{2} 2^{n/2}, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

Proof. Consider one reflection. If its axis passes through a bead, that bead is mapped to itself, and invariance under complementation would require its bit to equal its own complement, which is impossible; hence such a reflection fixes no configuration. If the axis passes through two opposite gaps between beads, it pairs the n beads into $n/2$ two-element orbits, each consisting of a bead and its complementary partner; a fixed configuration is obtained by choosing the bit of one bead in each pair freely, giving $2^{n/2}$ fixed configurations.

For n odd, every reflection axis passes through exactly one bead, so all n reflections fix nothing and $R(n) = 0$. For n even, exactly $n/2$ axes pass through two opposite beads (fixing nothing) and $n/2$ axes pass through two opposite gaps (fixing $2^{n/2}$ each), whence $R(n) = \frac{n}{2} 2^{n/2}$. $\square$

The count $R(n) = \frac{n}{2} 2^{n/2}$ was confirmed by direct enumeration of fixed points over all 2^n configurations for every even $n \leq 40$.

4.2 The closed form

Theorem 4.2. *For every $n \geq 1$, the number of oriented cyclic graphs on n edges up to D_n is*

$$a(n) = \frac{1}{2n} \left[\sum_{d|n} \varphi\left(\frac{n}{d}\right) 2^d + [2 \mid n] \frac{n}{2} 2^{n/2} \right], \quad (4.1)$$

where φ is Euler's totient and $[\cdot]$ is the Iverson bracket. The right-hand side is an integer for all n .

Proof. By Lemma 3.1 we count D_n -orbits of all of $\{0, 1\}^n$. Burnside's lemma gives $a(n) = \frac{1}{2n} (\sum_{k=0}^{n-1} |\text{Fix}(r^k)| + R(n))$. For the rotations, $|\text{Fix}(r^k)| = \text{tr}(M^{\text{gcd}(n,k)}) = 2^{\text{gcd}(n,k)}$ by Proposition 3.3 and Corollary 3.5; grouping the n rotations according to $d = \text{gcd}(n, k)$, of which there are $\varphi(n/d)$ for each divisor $d \mid n$, yields $\sum_k |\text{Fix}(r^k)| = \sum_{d|n} \varphi(n/d) 2^d$. Substituting Lemma 4.1 for $R(n)$ gives (4.1). Integrality follows because the left-hand side counts orbits. $\square$

Corollary 4.3. *Writing $A(n) = \frac{1}{n} \sum_{d|n} \varphi(n/d) 2^d$ for the number of binary necklaces on n beads (OEIS A000031),*

$$a(n) = \frac{1}{2} A(n) + [2 \mid n] 2^{n/2-2}.$$

This is the sequence A053656, whose n th term is the number of bracelets on n beads of two colours in which the colours are exchanged when the bracelet is turned over [1, 3].

The first values are

$$a(1), \dots, a(15) = 1, 2, 2, 4, 4, 9, 10, 22, 30, 62, 94, 192, 316, 623, 1096,$$

in agreement with A053656.

4.3 Asymptotics

Corollary 4.4. *As $n \rightarrow \infty$,*

$$a(n) = \frac{2^n}{2n} \left(1 + \mathcal{O}(2^{-n/2})\right) \sim \frac{2^n}{2n}.$$

Proof. In (4.1) the divisor $d = n$ contributes $\varphi(1)2^n = 2^n$; every other divisor satisfies $d \leq n/2$, so the remaining rotational terms sum to $\mathcal{O}(n 2^{n/2})$, and the reflection term is likewise $\mathcal{O}(n 2^{n/2})$. Dividing by $2n$ gives the claim. $\square$

Figure 3a displays the ratio $a(n) 2n/2^n$, which approaches 1 from above on even n and lies essentially on 1 for odd n ; Figure 3b shows that the number of decimal digits of $a(n)$ follows the predicted slope $n \log_{10} 2$.

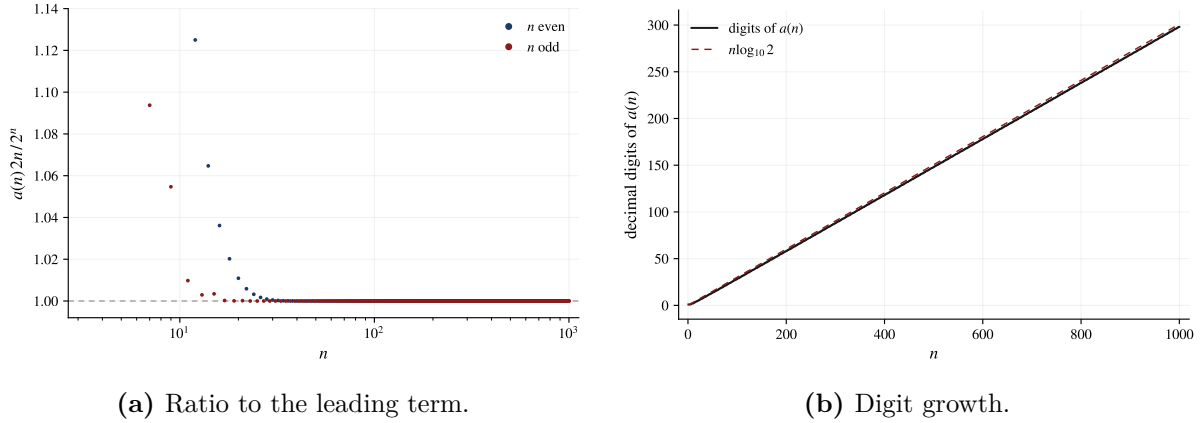


Figure 3: Left: $a(n) 2n/2^n \rightarrow 1$; the even and odd subsequences converge from opposite sides, the reflection term accounting for the $\mathcal{O}(2^{-n/2})$ excess on even n . Right: decimal digits of $a(n)$ against the line $n \log_{10} 2$.

5 Verification

The closed form (4.1) was checked against two independent computations.

- *Exhaustive enumeration* for $1 \leq n \leq 20$: all 2^n configurations were generated, reduced to canonical form under D_n , and counted. This is feasible only for small n ; at $n = 20$ it processes 2^{20} configurations.
- *Burnside evaluation* for $1 \leq n \leq 1000$: the divisor sum $\sum_{d|n} \varphi(n/d) \text{tr}(M^d)$ was formed with $\text{tr}(M^d)$ computed by binary exponentiation of M , and combined with the reflection term.

All three methods agree at every tested n ; no discrepancy was observed. Table 1 lists the first fifteen values.

Table 1: The first fifteen values of $a(n)$, obtained identically by exhaustive enumeration, Burnside evaluation, and the closed form (4.1). The sequence is OEIS A053656.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$a(n)$	1	2	2	4	4	9	10	22	30	62	94	192	316	623	1096

6 Computational complexity

We compare three algorithms for computing $a(n)$ in exact integer arithmetic.

Exhaustive enumeration. Generating and canonicalising all 2^n configurations costs $\Theta(n 2^n)$ time and, in the worst case, $\Theta(2^n)$ space. This is intractable beyond $n \approx 25$.

Burnside with matrix powers. Evaluating $\sum_{d|n} \varphi(n/d) \operatorname{tr}(M^d)$ requires $\mathcal{O}(d(n) \log n)$ matrix multiplications of the fixed 6×6 matrix M , where $d(n)$ is the number of divisors of n . The dominant cost is the arithmetic on the integers $\operatorname{tr}(M^d) \leq 2^d$.

Closed form. Equation (4.1) avoids matrix powers entirely, replacing $\operatorname{tr}(M^d)$ by 2^d . It requires enumerating the divisors of n in $\mathcal{O}(\sqrt{n})$ time, a totient evaluation per divisor, and the formation of the integer 2^n , which dominates.

Let $M(k)$ denote the cost of multiplying two k -bit integers; the reference implementation uses Python’s built-in arithmetic, for which $M(k) = \mathcal{O}(k^{1.585})$ (Karatsuba). The integer 2^n has n bits, so the closed form runs in $\mathcal{O}(\sqrt{n} + M(n) \log n)$ time.

Table 2 reports measured timings for the closed form, and Figure 4 plots the three algorithms together. The separation is stark: exhaustive enumeration exceeds eleven seconds at $n = 20$, whereas the closed form returns $a(3 \cdot 10^6)$, an integer of 903,084 decimal digits, in 36 milliseconds.

Table 2: Wall-clock timings of the closed form on a single core (CPython 3.14). “Digits” is the number of decimal digits of $a(n)$.

n	Digits of $a(n)$	Time
10^3	298	<1 ms
10^4	3,006	1 ms
10^5	30,098	1 ms
10^6	301,024	13 ms
$2 \cdot 10^6$	602,054	24 ms
$3 \cdot 10^6$	903,084	36 ms

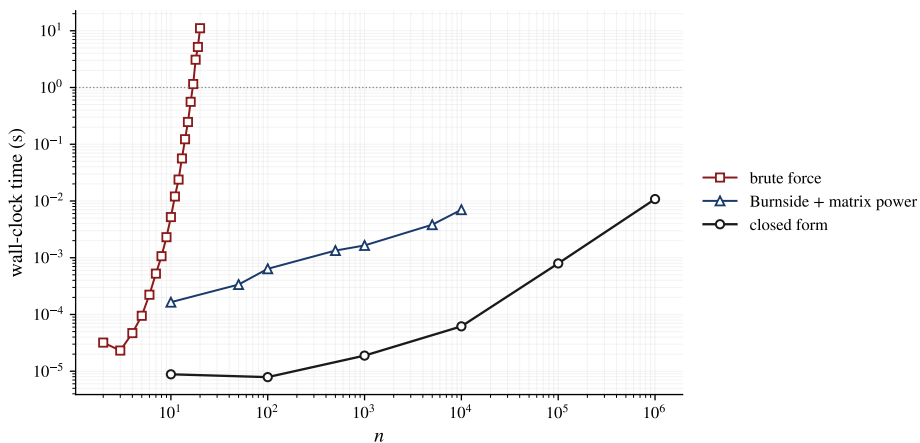


Figure 4: Wall-clock time to compute $a(n)$ exactly by three methods, on identical hardware. Exhaustive enumeration grows exponentially and leaves the frame near $n = 20$; the Burnside and closed-form methods are polynomial, the latter with the smaller constant.

7 Remarks and extensions

Primitive orbits. An orbit is *primitive* if its representative is not a shorter block repeated. By Möbius inversion the number of primitive orbits is $\sum_{d|n} \mu(d) a(n/d)$, computable at the same asymptotic cost as (4.1), since the required values $a(n/d)$ range over divisors of n .

Interpretations. By Corollary 4.3, $a(n)$ also counts binary bracelets whose colours are exchanged under reversal, and (equivalently) certain frieze patterns generated by an asymmetric motif [3]. These identifications place the present construction within the well-studied theory of complementing necklaces.

Stronger constraints. The transfer-matrix framework accommodates genuine local constraints, for instance forbidding a fixed cyclic subword, by enlarging the state set. Whether the resulting count again admits a closed form depends on the factorisation of the corresponding characteristic polynomial.

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